



Magnetospheric imaging with low-energy neutral atoms

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Communicated by Gian-Carlo Rota, July 15, 1991

ABSTRACT Global imaging of the magnetospheric charged particle population can be achieved by remote measurement of the neutral atoms produced when magnetospheric ions undergo charge exchange with cold exospheric neutral atoms. Previously suggested energetic neutral atom imagers were only able to measure neutral atoms with energies typically greater than several tens of keV. A laboratory prototype has been built and tested for a different type of space plasma neutral imaging instrument, which allows neutral atoms to be imaged down to <1 keV. Such low-energy measurements provide greater sensitivity for imaging the terrestrial magnetosphere and allow the bulk of the magnetospheric ion distribution, typically centered below 10 keV, to be observed rather than just the high-energy tail of the distribution. The low-energy neutral atom measurements are made possible by utilizing charge state modifications that occur when an initially neutral atom passes through an ultrathin carbon foil. Oxygen, for example, is highly electronegative, and for energies of ≈ 10 –30 keV, the O^+ yield is $\approx 30\%$, essentially independent of the charge state of the incident oxygen atom. These ions are energy per charge analyzed, and the UV background is rejected by using an electrostatic analyzer. Imaging of other ion species, such as hydrogen, could also be accomplished by using ultrathin foil-induced charge state modifications. The technique described in this paper provides a method for imaging charge exchange neutrals from the terrestrial magnetosphere and would also have applications for similar imaging in other planetary or cometary environs. The Inner Magnetosphere Imaging Mission, which the National Aeronautics and Space Administration is presently considering, would provide a nearly ideal platform for low-energy neutral atom imaging, and such measurements would substantially enhance the scientific yield of this mission.

Present knowledge of the terrestrial magnetosphere has been derived over the past three decades from single point spacecraft measurements of the local particle and field environments. Although much information about the global attributes of the magnetosphere has been gleaned through such observations, simultaneous measurements throughout the magnetosphere are still missing. This type of simultaneous, global information is important for understanding the overall structure and dynamics of the magnetosphere and can realistically be achieved only through remote sensing techniques such as imaging of the plasma population.

The National Aeronautics and Space Administration's Space Physics Division is presently studying a new Inner Magnetosphere Imaging Mission. This mission is designed to observe remotely the terrestrial inner magnetosphere including the ring current, plasmasphere, and inner edges of the plasma sheet as well as the auroral zone (1). Energetic neutral atom (ENA) imaging (2–6), utilizing neutrals with energies above several tens of keV, is perhaps the most promising measurement technique for observing and studying the global

plasma configuration and dynamics of a significant portion of the inner magnetosphere.

Energetic neutral atoms are produced from charge exchange of hot magnetospheric ions with cold gas atoms from the terrestrial neutral corona. Once an ion becomes neutralized, it is no longer affected by the local electric and magnetic fields, and it travels ballistically onward with the velocity that it had at the instant of charge exchange. Consequently, such neutrals are found continuously radiating outward from the inner magnetosphere, and remote measurements of these neutrals provide line-of-sight integrated images of the magnetospheric plasma populations. These images of the neutral atom emission can be deconvolved to provide the distributions of ion source populations (2, 3). If such images are made with energy- and/or mass-resolving instrumentation, then so too are the measurements of the source populations.

Although several viable techniques have been developed for imaging energetic neutral atoms (4–6), these measurements are not able to address the bulk of the inner magnetosphere and plasma sheet populations, which are populated by ions with energies of typically less than several tens of keV. In addition, fluxes of low-energy charge exchange neutrals are somewhat greater than at high energies, allowing measurements to be made further out in the magnetosphere. The next section provides a calculation of the storm-time low-energy neutral fluxes radiating from the inner magnetosphere.

The primary problem with the direct measurement of lower energy magnetospheric neutral atoms is the presence of a very large flux of Ly- α extreme ultraviolet (EUV) photons ($>10^{10}$ cm $^{-2}$ s $^{-1}$) scattered from the geocorona (7). EUV photons have sufficient energy to stimulate low-energy particle detectors, such as microchannel plates (MCPs), and therefore must be effectively rejected from an imaging detector. ENA instruments typically rely on foils to attenuate the EUV. Foils of thickness >15 μ g/cm 2 provide sufficient EUV rejection to make photon-induced background tolerable (4, 7). Unfortunately, such thick foils also cause severe energy straggling and angular scattering of the transmitted ENAs. For low-energy neutral atoms (LENAs), with energies less than several tens of keV, such thick foils make meaningful measurements nearly impossible.

In this paper we describe a technique for imaging neutral atoms that does not rely on the entrance foil to screen out incoming UV photons. Rather, an ultrathin (<1 μ g/cm 2) foil is used to modify the charge state of a fraction of the incident neutral atoms. Newly formed ions are subsequently energy per charge (E/q) analyzed with an electrostatic analyzer, which also effectively rejects the EUV photons. This technique differs importantly from previously suggested thin foil/multiple coincidence measurements (8) in that we reject the large EUV flux prior to detection of the ions, and therefore it is not necessary to extract the small signal from very large background counting rates. To the best of our knowledge, only a single sounding rocket experiment has

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Abbreviations: EUV, extreme ultraviolet; E/q , energy per charge; ENA, energetic neutral atom; MCP, multichannel plate; LENA, low-energy neutral atom; TOF, time of flight; R_E , earth radii.

ever attempted to measure LENAs in space (9, 10). Although several fundamental aspects of the technique used by these authors were similar to that described here, their measurements were made over a single, integrated field of view, and their instrument was not an imaging detector.

The technique described here makes it possible to extend the energy of neutral imaging downward from several tens of keV to <1 keV, through the range of energies where the magnetospheric charge exchange neutral fluxes are greatest. This technique could provide (i) previously unattainable sensitivities for neutral imaging with a reasonably sized space instrument, (ii) information about the bulk of the magnetospheric ion population over much of the inner magnetosphere, and (iii) observations about otherwise unobservable magnetospheric structures such as the near-earth plasma sheet.

Charge-Exchange Neutral Fluxes

Magnetospheric neutral atoms are produced by charge exchange between magnetospheric ions and atoms from the neutral exosphere, which, beyond a few thousand kilometers altitude, is dominated by hydrogen. The Dynamics Explorer 1 best fit to the atomic hydrogen density profile (11) is given by

$$n_H(R) = 3300 e^{(-R/1.6)} \text{ cm}^{-3}, \quad [1]$$

where $n_H(R)$ is the neutral hydrogen density as a function of geocentric distance in earth radii (R_E).

The unidirectional flux of charge exchange neutrals of species i , $f_i(E)$, is then given by (4) the integral along the line of sight, l ,

$$f_i(E) = \sigma_{iH}(E) \int_l J_i(l, E) n_H(l) dl, \quad [2]$$

where $\sigma_{iH}(E)$ denotes the energy-dependent charge exchange cross-sections for various ion species with neutral hydrogen, $\sigma(i^+ + H \rightarrow i + H^+)$. The flux of magnetospheric ions of species i , $J_i(l, E)$, is in general a function of energy, pitch angle, and location in the magnetosphere. Eq. 2 is independent of observing location as long as the source region uniformly fills the instrument's field of view, since the neutral flux from any particular plasma element drops off as the square of the distance while the number of emitting elements observed increases as the square of the distance.

The dominant magnetospheric ions are H^+ and O^+ , with relative abundances that are both highly variable and dependent on geomagnetic activity. While the quiet-time magnetosphere is dominated by H^+ , O^+ can become comparable during storm times (12). Low-energy neutral imaging of magnetospheric hydrogen is complicated by a background due to the high fluxes of hot magnetosheath hydrogen ions and the rather slow rate at which the exospheric neutral density drops off beyond $\approx 6R_E$ (11). Imaging of the magnetosheath ion population, which resides nearly entirely below several keV, is scientifically interesting in its own right. However, there is an ambiguity in measuring the low-energy neutral fluxes from the magnetosheath and the portion of the magnetospheric hydrogen population that resides below several keV. The specifics of separating these populations are complicated and highly dependent on the particular spacecraft orbit.

For the purposes of describing our technique, this paper concentrates on the simpler case of imaging low-energy neutral oxygen, which is of almost purely terrestrial origin and is essentially absent outside the magnetopause. Fig. 1 displays energy spectra for fluxes of various storm-time ring

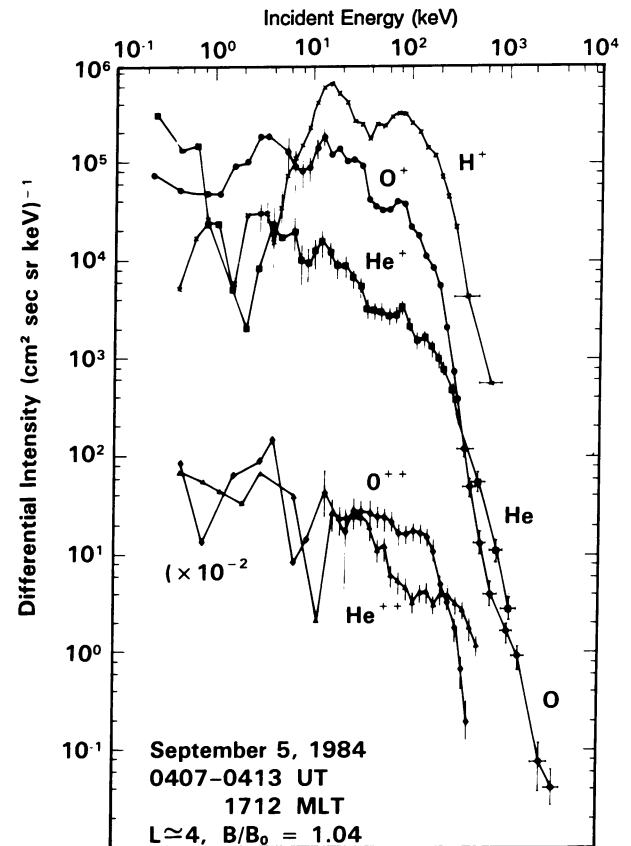


FIG. 1. Energy spectra of various storm-time ring current ion fluxes measured by the AMPTE Charge Composition Explorer spacecraft. [Reproduced from ref. 12 (copyright American Geophysical Union).]

current ions measured by the AMPTE Charge Composition Explorer spacecraft (12). It is clear from Fig. 1 that the highest fluxes of ring current hydrogen and oxygen are found at energies less than several tens of keV.

Although deconvolution of the spatial dependence of this ion flux is perhaps the most difficult aspect of analyzing neutral imaging data (2, 3), simple approximations can be employed for the purposes of determining approximate flux levels. Based on the observed radial dependence of the number density of ring current ions with energies greater than 5 keV (12), we adopt an empirical relation for the equatorially mirroring flux of ring current oxygen $J_{O^+}(R, E)$:

$$J_{O^+}(R, E) = J_{O^+}(4, E) e^{-0.46(R-4)}, \quad [3]$$

where $J_{O^+}(4, E)$ is determined from the observed storm-time spectrum in Fig. 1. The energy-dependent charge exchange cross-section for $O^+ + H \rightarrow O + H^+$ is given by Phaneuf *et al.* (13). Substituting the charge exchange cross-sections and Eqs. 1 and 3 into Eq. 2, and numerically integrating this relation inside of $9R_E$ along various lines of sight in the equatorial plane, we calculate the approximate fluxes of neutral oxygen. Fig. 2 displays these estimated neutral oxygen fluxes as a function of energy for several different lines of sight, characterized by their closest approach distances to the earth, L_0 . These storm-time neutral oxygen fluxes peak at ≈ 6 keV and drop sharply at energies above ≈ 40 keV.

There are several advantages to measuring the neutral oxygen fluxes displayed in Fig. 2 down to low energies. First, the total integrated flux between 1 and 40 keV is about 3 times greater than the integrated flux above 40 keV. For $L_0 = 4R_E$, for example, these integrated storm-time fluxes are $\approx 2600 \text{ cm}^{-2}\text{sec}^{-1}\text{steradian}^{-1}$ and $\approx 900 \text{ cm}^{-2}\text{sec}^{-1}\text{steradian}^{-1}$ for

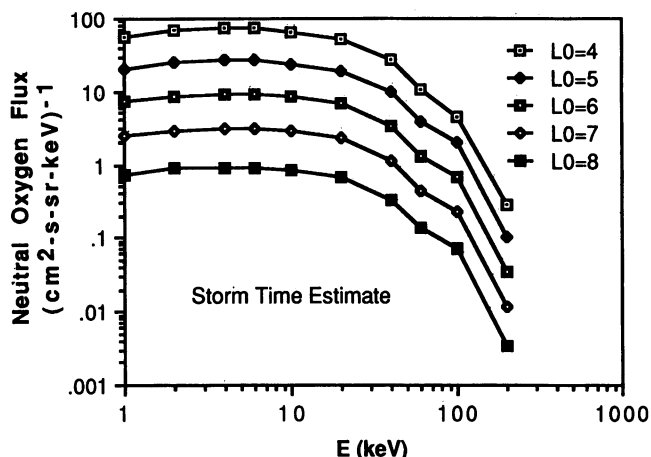


FIG. 2. Calculated neutral oxygen fluxes as a function of energy for equatorial lines of sight, which have closest approach distances to the earth of $L_0 = 4$ to $L_0 = 8 R_E$. These storm-time neutral oxygen fluxes peak below 10 keV and drop sharply at higher energies.

energies below and above 40 keV, respectively. In addition, the assumed radial dependence of the ion flux (Eq. 3) is most accurate for the lower energy ions, whereas more energetic ion fluxes probably drop off even more quickly with radial distance, further reducing the neutral fluxes at high energies. Finally, since the decay of the ring current occurs more rapidly at energies >20 keV (12), low-energy measurements of magnetospheric oxygen will become increasingly important as observations progress into the recovery phase of the storm.

The Interaction of Neutral Atoms with Thin Foils

When atoms traverse thin foils, they interact in several characteristic ways, which depend importantly on their energy per nucleon (speed) and on the electronegativities of both the atoms and the foil. These effects include modification of the atoms' charge states, angular scattering, and energy straggling. In addition, atoms often eject electrons from the back surface of the foil as they exit. This phenomenon has been used to provide start timing for time-of-flight (TOF) mass spectrometers in numerous space applications (14).

Evaporated carbon is the most commonly used material for thin foils in space applications. Nominal thicknesses from $0.1 \mu\text{g}/\text{cm}^2$ ($\approx 20 \text{ \AA}$) to several tens of $\mu\text{g}/\text{cm}^2$ (thousands of $\text{\AA}$) are commercially available. With sufficient care, even the thinnest foils can be transferred to high transmittance grids; we have transferred and tested carbon foils in the laboratory with nominal thicknesses down to $0.1 \mu\text{g}/\text{cm}^2$ and have successfully flown in space foils with nominal thicknesses of $0.5 \mu\text{g}/\text{cm}^2$ (15).

The method described here for measuring low-energy neutral atoms is based on the modification of charge state that atoms undergo when they pass through a thin foil. In particular, low-energy atoms (with a velocity less than approximately the Bohr velocity) can be considered to be part of the bulk foil and rapidly come into charge state equilibrium. Consequently, the initial charge states of the incident atoms are unimportant, and the distribution of exit charge states is mainly dependent on the physical properties of the atoms and foil. Recent experiments show that O^+ and O^{+7} from ≈ 0.8 to ≈ 1.5 keV per atomic mass unit incident on a $2 \mu\text{g}/\text{cm}^2$ carbon foil produce almost identical exit charge state distributions, indicating that the exit charge state of a particle is essentially independent of its initial charge state (16).

Oxygen is strongly electronegative and readily removes electrons from a carbon foil. For oxygen with energies of 12–32 keV, the approximate O^+ yield varies from 6% to 13%, the O^0 yield varies from 70% to 59%, and the O^- yield varies from 30% to 28% (16, 17). Most other species exit the foil with distributions of charge states that are far more dominated by the neutral component (typically $\approx 90\%$ neutral). Since such a large fraction of an initially neutral oxygen beam exits the foil negatively charged and since no other magnetospherically abundant atoms exit the foil with a substantial negative yield, an instrument that analyzes negative ions subsequent to passing through a thin carbon foil provides a unique signature of incident oxygen atoms.

In addition to the conversion of charge states in a thin foil, energies are reduced (straggling) and atoms are deflected (scattering). Unlike charge state conversion, which is essentially independent of foil thickness, the straggling and scattering scale with the thickness of the foil. Both of these effects are generally undesirable in imaging instruments, and minimizing their importance is crucial in designing foil-based LENA imaging instrumentation.

LENA Imaging Instrument Design and Prototype Testing

Whereas previously suggested ENA imagers have utilized rather thick carbon foils to reject UV light (and provide start-timing electrons in some cases), the technique described here relies on the charge state conversion of atoms transiting a very thin carbon foil. At thicknesses $<1 \mu\text{g}/\text{cm}^2$, very little UV rejection is achieved by the foils. However, since $\approx 30\%$ of the incident oxygen exits the foil as O^- , subsequent E/q analysis with an electrostatic analyzer can provide excellent UV rejection as well as energy analysis.

Fig. 3 is a schematic diagram of one version of our LENA imaging instrument. This design is one of many appropriate for imaging from a spinning spacecraft. Incident neutral atoms enter the device through the collimator, which defines the angular acceptance of the sensor and additionally sweeps out charged particles (ions and electrons) from the local spacecraft environment with an imposed electric field. The neutral atoms pass unperturbed through the collimator and encounter the thin carbon foil (nominally $0.2 \mu\text{g}/\text{cm}^2$ in our prototype testing), which is supported on a high transmission

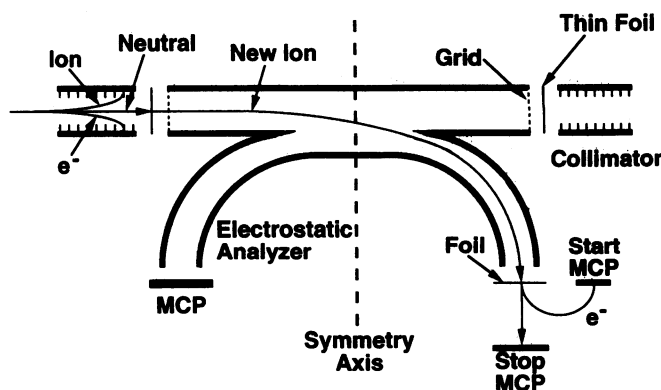


FIG. 3. Schematic diagram of a cross-section of one possible LENA imaging sensor appropriate for a spinning spacecraft. Neutral atoms are collimated, and incident electrons and ions are swept out, by the collimator section (which has high voltage across it). A fraction of the transmitted LENAs become ionized in the thin foil. These ions are postaccelerated by a high voltage applied to the foil and are subsequently E/q analyzed in the electrostatic analyzer. Ions are either detected directly on the MCPs (left side) or with a TOF mass spectrometer (right side), which determines their mass and substantially reduces the instrumental noise background by providing a coincidence measurement.

grid. Postacceleration, achieved by applying negative high voltage to the foils, energizes the exiting negative ions and allows for focusing of the ion trajectories. The trajectories of lower energy atoms, which typically undergo greater angular scattering, are more easily focused.

A spherical top hat (18) or toroidal electrostatic analyzer (19), with a grounded grid at its entrance, E/q analyzes the postaccelerated negative ions. By grooving and blackening the plates, essentially all of the UV can be rejected, as is typically done in space-borne plasma analyzers (20). The specific configuration of the collimator and electrostatic analyzer depends both on the specific orbital and spacecraft attributes and on the intrinsic energy and angular resolutions required. Detection of the E/q-analyzed ions could either be achieved with MCP detectors (shown on the left side of the figure) or with a mass spectrometer section.

The right side of Fig. 3 displays a TOF mass spectrometer (14, 15) behind the electrostatic analyzer. This device provides an additional coincidence measurement, which effectively rejects most sources of instrumental noise in the measurements and greatly improves the sensor's signal-to-noise ratio. Consequently, it is possible to integrate over long intervals in order to obtain maximum sensitivity. Electrons created by the neutral atoms and EUV photons interacting with the foil can be effectively removed either magnetically or by avoiding their postaccelerated energy with the electrostatic analyzer sweep.

To examine this technique, we have constructed and tested a prototype sensor. The design is essentially that shown in Fig. 3 except that only a single look direction is populated with a carbon foil. After E/q analysis, the O^- ions were collected on an imaging MCP detector. The image was monitored in order to assure that the observed count rates came almost entirely from the analyzer exit gap. Count rates were collected in a computer-based multichannel analyzer as a function of electrostatic analyzer voltage.

Fig. 4 displays the results for monoenergetic 1-keV, 5-keV, and 20-keV oxygen beams. The amplitudes of the three peaks have been scaled by factors of 1.05, 0.305, and 1.00, respectively, to make the peak shapes more easily comparable. Since monoenergetic neutral oxygen beams are very hard to produce at these energies and since the exit charge state distribution is essentially independent of the entrance distribution (as discussed above), we have substituted an incident O^+ beam in this testing.

The full widths at half-maximum of the two higher energy peaks are $\approx 13\%$, which is the intrinsic energy resolution of the toroidal electrostatic analyzer used for this prototype. The 1-keV peak has a full width at half-maximum of $\approx 31\%$, indicating that energy straggling in the foil has become an

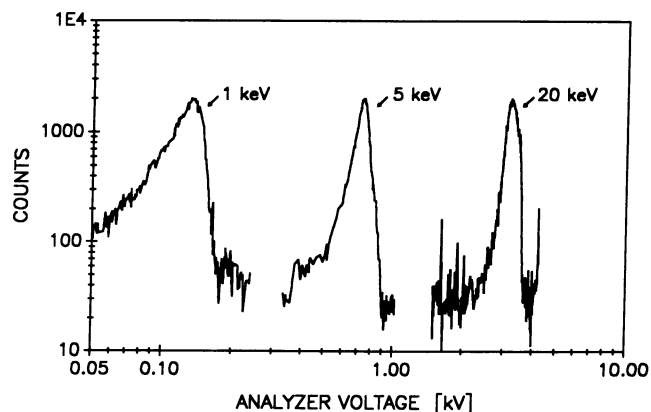


FIG. 4. LENA-type measurements obtained with our prototype sensor using 1-, 5-, and 20-keV incident oxygen beams and no postacceleration.

important effect at this low energy. For simplicity, the three data runs shown in Fig. 4 were made with no postacceleration voltage on the foil. Postacceleration and ion focusing would be used in a flight instrument to increase the effective aperture of the sensor and enhance the detection efficiency for lower energy atoms, which undergo substantial angular scattering even in very thin foils.

Discussion

In this paper we have described a technique for imaging LENAs. The required instrumentation is based on well-developed plasma analyzer and mass spectrometer designs that can be readily achieved in compact and lightweight space sensors (14, 15, 18–20). The fundamental aspects of the device are (i) removal of incident ions with an electric field across an entrance collimator, (ii) charge state modification of some fraction of the incident neutral atoms in a thin foil, (iii) postacceleration and focusing of the resultant ions to increase instrument sensitivity, (iv) electrostatic analysis to reject EUV photons as well as to resolve the energy distribution of the incident neutrals, and (v) detection of the ions directly, or further mass analysis and noise rejection with a TOF section.

The example shown here utilized O^- ions, which provides a unique signature for oxygen in the magnetosphere. This technique, however, is also appropriate for the analysis of hydrogen and other magnetospheric charge exchange neutrals by analyzing positive ions from the foils. If positive ions are used, a mass spectrometer behind the electrostatic analyzer would be highly desirable in order to identify the various species observed. A TOF mass spectrometer, or other form of coincidence or anticoincidence measurement, is a desirable element of the sensor in any case, since it provides a measurement with a greatly improved signal-to-noise ratio.

For a spinning spacecraft, like the proposed Inner Magnetosphere Imaging Mission, the spherical or toroidal type top-hat analyzer provides a good combination of energy and angular resolution with high sensitivity. A lightweight, high-sensitivity toroidal analyzer has been well characterized by Young *et al.* (19). Its effective aperture area exceeds 2 cm^2 for a single look direction, and it has a non-energy-resolved instrumental geometric factor of $\approx 1.7 \text{ cm}^2\text{-steradian}$ for a $7.6^\circ \times 360^\circ$ field of view. Its intrinsic full width at half-maximum energy and angular resolutions are 18% and 7.6° , respectively.

The spacecraft orbit, spin rate, and other factors will obviously play a large role in dictating the particular specifications of the collimator and electrostatic analyzer design used. However, if an analyzer with the geometric factor given above were staring (nonspinning) at a region emitting the neutral fluxes calculated above, non-energy-resolved images with $7.5^\circ \times 7.5^\circ$ resolution ($\approx 1R_E$ at $9R_E$ distance) would be measured with ≈ 1000 counts per pixel in about 35 sec ($L_o = 4$). Even out at $L_o = 8$, >100 counts per pixel could be collected in 5 min. Making both differential energy measurements and measurements from spinning spacecraft will reduce the collection times for a given look direction and energy step. Fortunately, even larger geometric factors than that given above are achievable in reasonable-sized space instruments.

We gratefully acknowledge Don Mitchell for many useful discussions and suggestions. Laboratory testing of our prototype sensor was carried out with substantial support of Bob Baldonado, Danny Everett, Neal Lundgaard, Nick Olivas, and Dave Suszcynsky. This work was carried out under the auspices of the United States Department of Energy.

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